



Forward and Back

Quick facilitation notes & answer key

Goal

Students run programs that mix forward (add) and back (subtract) blocks and write one of their own, connecting sequential code to adding and subtracting within 10. This is reading/running and writing code for a mathematical situation (C3.1) with a predict-and-check step (C3.2).

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 so students can step forward and back to feel adding and taking away.

Run it (about 20 min)

1. I DO: run Code A — 'start at 1... forward 3 to 4... back 1 to 3.' Name it: forward adds, back takes away.
2. WE DO: Exercise 1 — Code A lands on 3. Exercise 2 — predict Code B, then run: forward 1 to 4, back 2 to 2.
3. YOU DO: Exercise 3 — from 2, use one forward and one back block to reach 4 (e.g. forward 3, back 1).
4. Connect: 'Forward is like adding and back is like subtracting — coding the number path is doing math.'

Answer key (modelled by the run-gate)

Ex 1 — Code A: 1 → forward 3 → 4 → back 1 → 3.

Ex 2 — Code B: 3 → forward 1 → 4 → back 2 → 2.

Ex 3 — start 2 → 4: e.g. forward 3, back 1 (net forward 2). Other mixes that net +2 also work, e.g. forward 2 then back 0 is not allowed (no back-0 block), so forward 3/back 1 is the clean pair.

Why it works: each block runs in order; forward adds its number and back subtracts its number, so the landing is start + forwards – backs (kept on the 0-5 path).

Watch for & differentiate

Common slip: treating a back block as forward — point the back arrow/word and step the wrong way once to contrast.

Simplify: do Codes A and B only; skip the build task.

Extend: ask for a program from 2 to 4 that uses a back block first (e.g. back 1 to 1, then forward 3 to 4).

Success indicator: the child lands Code A on 3 and Code B on 2 and writes a forward+back program that reaches 4.